

House Price Prediction – Project Summary

Objective →

The objective of this project was to develop a machine learning model to predict house prices using property features such as area, bedrooms, bathrooms, parking, location preference, and furnishing status.

Dataset Overview →

The dataset contained 545 records and 13 features. No missing values were found, so no data imputation was required. Categorical features were converted into numerical form using one-hot encoding before model training.

Models and Performance →

Model	MAE	RMSE	R ² Score
Linear Regression	970,043	1,324,507	0.653
Random Forest Regressor	1,021,546	1,400,566	0.612

The Linear Regression model performed better than the Random Forest Regressor, achieving a higher R² score and lower prediction errors.

Key Findings →

Feature importance analysis showed that the most influential factors affecting house prices were:

- Area (46.84%)
- Bathrooms (15.15%)
- Air Conditioning (6.27%)
- Parking Availability (5.75%)
- Number of Stories (5.71%)

Among all features, area had the strongest impact on house prices.

Conclusion →

The analysis showed that house size is the most important factor in determining property value. The Linear Regression model explained approximately 65% of the variation in house prices, indicating reasonable predictive performance. Amenities such as air conditioning, parking, and additional bathrooms also contributed positively to property prices.

Recommendation →

Real estate businesses should focus on property size, bathrooms, parking facilities, and modern amenities when estimating house prices and marketing properties to potential buyers.